

Enzyme Bench

Two inhibitors, one question: which of them can substrate out-vote, and which of $V_{\max}$ and K_m moves when it does.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/enzyme-bench/

1 Before you open anything

Answer from what you already know. Being wrong here costs nothing — the point is to have a committed number to argue with later.

PREDICTION

An enzyme is inhibited, and you respond by flooding the tube with substrate — enough to saturate the enzyme completely. For each inhibitor type, what percentage of the *uninhibited* $V_{\max}$ do you expect to get back?

Competitive inhibitor: _____ % Noncompetitive inhibitor: _____ %

Write one or two sentences saying what reasoning produced those two numbers.

2 The uninhibited enzyme

Before anything is inhibited, you need to know what normal looks like. Open the demonstration and press **Reset all**.

1. Confirm the starting conditions: $[E]_{\text{total}} = 100 \text{ nM}$, **Temperature 37.0 °C**, **pH 7.0**, and both inhibitor checkboxes clear. The **Activity vs 37 °C, pH 7** tile should read 100%.
2. Write down what the $V_{\max}$ (**apparent**) and K_m (**apparent**) tiles read right now.
3. Drag the **[S]** slider until its readout shows each value in the table, and record **Rate v now** and **Share of $V_{\max}$** .

$V_{\max} = \text{_____ } \mu\text{mol L}^{-1} \text{ min}^{-1}$ $K_m = \text{_____ mM}$

[S] readout	Rate v now ($\mu\text{mol L}^{-1} \text{min}^{-1}$)	Share of V_{max}
0.500 mM		
2.00 mM		
6.00 mM		
20.0 mM		
100 mM		

a. One row reads exactly 50.0% of V_{max} . Which [S] is it, and what is that concentration called?

b. Going from 20.0 mM to 100 mM multiplies [S] by five. By what factor does the rate increase? Explain, using the shape of the curve, why buying that much substrate is a poor investment.

c. Set $[E]_{\text{total}}$ to 50 nM. Record both headline tiles, then return the slider to 100 nM.

$V_{\text{max}} = \text{-----}$ $K_m = \text{-----}$

Which one changed, and why does the other one have no reason to?

3 What each inhibitor actually moves

Now vary one thing at a time: the dose of one inhibitor, with [S] held fixed. Keep [S] at 2.00 mM for both tables — press **Reset all** between them if the slider drifts.

1. Tick **Competitive inhibitor**. Its slider unlocks; it runs 0 to 10.0 mM in steps of 0.1, and $K_i = 2.0$ mM.
2. Set **[I] competitive** to each dose in the table and record the two headline tiles and **Rate v now**. For the 0 mM row, simply leave the checkbox unticked.
3. Untick it, tick **Noncompetitive inhibitor** instead, and repeat.

Competitive — [S] fixed at 2.00 mM

[I] competitive	$V_{\max}$ (apparent)	K_m (apparent)	Rate v now
0 mM			
2.0 mM			
4.0 mM			
6.0 mM			
10.0 mM			

Noncompetitive — [S] fixed at 2.00 mM

[I] noncompetitive	$V_{\max}$ (apparent)	K_m (apparent)	Rate v now
0 mM			
2.0 mM			
4.0 mM			
6.0 mM			
10.0 mM			

a. In the competitive table, K_m at $[I] = 10.0$ mM is how many times K_m at $[I] = 0$? Show that $K_{m,app} = K_m(1 + [I]/K_i)$ gives the same factor.

b. State the rule in your own words: one sentence for what a competitive inhibitor does to $V_{\max}$ and K_m , one sentence for the noncompetitive one.

c. Press the **Competitive** preset button — it sets that inhibitor to 10.0 mM and clears the other — then press **Flood with substrate** and read the tile labelled **Rate at saturating [S] (500 mM), as a share of the uninhibited $V_{\max}$** . Now press **Noncompetitive** and read the same tile.

Competitive: _____ % Noncompetitive: _____ %

Compare both numbers with your prediction in Part 1. If either was off, say specifically which assumption produced it.

The competitive figure is not quite 100%, and it never will be at any finite [S]. The curve approaches the uninhibited $V_{\max}$ as an asymptote — 500 mM is close enough to make the point, not close enough to reach it.

4 Using it the way a question would

A single rate measurement is not evidence of anything. Here are two conditions that are indistinguishable at one substrate concentration.

- 1. Press **Reset all**. Confirm [S] reads 2.00 mM.
- 2. **Condition A:** tick **Competitive inhibitor** and set it to **6.0 mM**. Record the row.
- 3. **Condition B:** untick it, tick **Noncompetitive inhibitor** and set it to **3.0 mM**. Record the row.
- 4. For each condition, also read the saturating-[S] rescue tile.

Condition (at [S] = 2.00 mM)	Rate v now	Share of $V_{\max}$	$V_{\max}$ (app)	K_m (app)	Rescue at 500 mM
A – competitive 6.0 mM					
B – noncompetitive 3.0 mM					

a. The **Rate v now** column is identical for A and B. Name two readouts on this page that tell them apart anyway, and say what each one shows.

b. A lab has one enzyme sample and can only afford a single assay. What would you tell them to change so that one further measurement distinguishes A from B? Justify it in one sentence.

c. Working backwards. You want the K_m (**apparent**) tile to read exactly **10.00 mM** using the competitive inhibitor alone. Calculate the dose first from $K_{m,app} = K_m(1 + [I]/K_i)$, then set the slider and check.

Calculated $[I] =$ _____ mM Tile reads: _____ mM

d. Press **Noncompetitive**, note V_{max} , then press **Neither** and lower $[E]_{total}$ until V_{max} matches. Record the enzyme concentration you needed, and explain what that equivalence says about how a noncompetitive inhibitor works.

$[E]_{total} =$ _____ nM

5 Solve these without the demonstration

Close the page or look away. Show your working. Then reopen it, reproduce each situation on the sliders, and write down where any disagreement came from. Unless a problem says otherwise, use $V_{\max} = 100 \mu\text{mol L}^{-1} \text{ min}^{-1}$, $K_m = 2.00 \text{ mM}$, and $K_i = 2.0 \text{ mM}$ for both inhibitors.

1. Calculate the initial rate at $[S] = 3.00 \text{ mM}$ with no inhibitor present, and express it as a percentage of $V_{\max}$.

2. A competitive inhibitor is added at $[I] = 5.0 \text{ mM}$. Predict $K_{m,\text{app}}$, then predict the initial rate at $[S] = 4.00 \text{ mM}$.

3. A double-reciprocal plot of two conditions gives two straight lines. Both cross the vertical axis at $1/v = 0.0100 \text{ L min } \mu\text{mol}^{-1}$. One crosses the horizontal axis at -0.500 mM^{-1} , the other at -0.125 mM^{-1} . Identify the inhibitor type, find $V_{\max}$ and both K_m values, and calculate $[I]$.

4. A noncompetitive inhibitor is present at a concentration that halves the rate at every $[S]$. State that concentration. Then explain, in terms of what the inhibitor does to individual enzyme molecules, why a solution containing half as much enzyme and no inhibitor at all would give an identical v -versus- $[S]$ curve.

6 On your own

Nothing here is graded, and the demonstration does considerably more than this worksheet used.

- Drag **Temperature** across its whole range and watch the left-hand environment plot. Find the two temperatures at which activity is half-maximal. The peak is not halfway between them — work out which of the two effects, the Arrhenius rise or the loss of folded enzyme, makes the curve asymmetric.
- Take the temperature to 55 °C, then bring it back to 37 °C. The rate returns. Read the *Honest limits* panel and say what would have to be added to the model for that recovery to stop happening.
- Press **Pepsin, opt 2.0**, then drag pH back to 7.0 and watch the **Activity vs 37 °C, pH 7** tile. Do the same starting from **Trypsin, opt 8.0**. Explain what this predicts about where in the body each enzyme can work.
- Switch on both inhibitors at once and open the **1/v vs 1/[S]** tab. Both intercepts move. Predict where each intercept lands before you look.